26900068 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBOT300 - PLANT BREEDING AND PLANT GENETIC RESOURCES

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Write notes on RRII. [U] / [CO1]
2. Write any two major achievements of IRRI in plant breeding. [U] / [CO1]
3. List one example of intergeneric hybridization. [K] / [CO2]
4. What is autopolyploidy? [K] / [CO2]
5. Define mutant. [K] / [CO2]
6. Define inbreeding depression. [K] / [CO3]
7. What is the fundamental principle of the bulk method in plant breeding? [U] / [CO3]
8. What is average-parent heterosis? [U] / [CO3]
9. Give the Binomial and family of Chathurappuli. [K] / [CO4]
10. What is germplasm? [K] / [CO4]

11. Define ethnobotany. [K] / [CO4]

12. State the main difference between in-situ and ex-situ conservation strategies. [U] / [CO5]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **8** questions

Each question carries **5** marks

13. Explain the roles and achievemnts of any two national centres of plant breeding. [U] / [CO1]

14. What you mean by Centres of origin? Explain Mediterranean centre and Abyssinian centre of origin. [U] / [CO1]

15. Discuss the advantages and disadvantages of mass selection. [U] / [CO2]

16. Write brief notes on the major plant introduction agencies in India. [U] / [CO2]

17. Write a short essay on plant acclimatization in plant breeding. [K] / [CO2]

18. Explain back cross method and its importance in improving established crop varieties. [U] / [CO3]

19. Compare the three hypotheses of heterosis in terms of their genetic basis and mechanism. [U] / [CO3]

20. Give the binomial, family and uses of following plants: a) Kattuloovikka b) Barnyard grass. [K] / [CO4]

21. Explain the land preparation and Soil requirements for the cultivation of Cow pea. [U] / [CO4]

22. Explain the ecological and genetic impacts of invasive plants on native endemic species. [U] / [CO5]

[5x8 = 40]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

23. What is clonal selection? Explain the procedure and advantages of clonal selection. [U] / [CO2]

24. Explain the climatic requirements, propagation methods, and manuring practices followed in the cultivation of Coffee and Ginger. [U] / [CO4]

[10x1 = 10]